F70LA Life Insurance Mathematics A (2025/26)
Tutorial 3 Solutions

1. (a)
$$ {}_{14}p_{37} = \frac{l_{51}}{l_{37}} = \frac{95964}{98217} = 0.97706. $$

(b)
$$ {}_{8}q_{60} = 1 - {}_{8}p_{60} = 1 - \frac{l_{68}}{l_{60}} = 1 - \frac{83150}{91732} = 0.09356. $$

2. For $t \leq 100 - x$,
$$ {}_{t}p_{x} = \frac{l_{x+t}}{l_x} = \frac{1,000,000\sqrt{100 - x - t}}{1,000,000\sqrt{100 - x}} = \sqrt{1 - \frac{t}{100 - x}}. $$

For $t > 100 - x$, ${}_{t}p_{x} = 0$ since (x) is dead within $100 - x$ years. Then

$$ \begin{aligned} \mathring{e}_{60} = \int_0^\infty {}_{t}p_{60}\, dt &= \int_0^{40} \left(1 - \frac{t}{40}\right)^{0.5} dt, \quad \text{since the limiting age is 100 years.} \\ &= \int_1^0 (-40)u^{0.5} du, \quad \text{by letting } u = 1 - \frac{t}{40}, \quad \text{so that } dt = -40du \\ &= 40 \int_0^1 u^{0.5} du = 40 \left[\frac{u^{1.5}}{1.5}\right]_0^1 = 26.67 \text{ years.} \end{aligned} $$

3. (a) ${}_{10}p_{30} = \frac{l_{40}}{l_{30}} = \frac{96500}{97645} = 0.98827.$
(b) ${}_{20}q_{30} = 1 - {}_{20}p_{30} = 1 - \frac{l_{50}}{l_{30}} = 1 - \frac{93925}{97645} = 0.038097.$
(c) ${}_{14}|q_{30} = \frac{l_{44} - l_{45}}{l_{30}} = \frac{d_{44}}{l_{30}} = \frac{230}{97645} = 0.002355.$

4. As ${}_{20}q_{65} = 1 - \frac{l_{85}}{l_{65}}$,

(a) using ELT15 (Males), ${}_{20}q_{65} = 1 - \frac{19756}{79293} = 0.75085.$
(b) using AM92 Ultimate, ${}_{20}q_{65} = 1 - \frac{3385.2479}{8821.2612} = 0.6162399.$
(c) using PMA92C20, ${}_{20}q_{65} = 1 - \frac{4892.878}{9647.797} = 0.49285.$

Why is (a) > (b) > (c)?

ELT15 is a general life table for the whole male population of England and Wales. People are included regardless of their health.

AM92 is only for assured lives, i.e. people who have a life assurance policy. There is a selection (by the life offices) taking place via a medical examination. People with bad health are less likely to be able to get insured.

PMA92C20 is a pension table projected to 2020. There might be some self-selection going on since the tables are constructed from insured lives (i.e. people who have a pension). The projection reflects increasing life expectancies in recent history due to progress in health care.

5. As ${}_{8}|_{7}q_{62} = \frac{l_{70}-l_{77}}{l_{62}}$,

   (a) using ELT15 (Females), ${}_{8}|_{7}q_{62} = \frac{79970-64561}{90132} = 0.17096$.

   (b) using AM92 Ultimate, ${}_{8}|_{7}q_{62} = \frac{8054.0544-6282.9803}{9129.7170} = 0.1939900$.

   (c) using PFA92C20, ${}_{8}|_{7}q_{62} = \frac{9392.621-8423.243}{9804.173} = 0.098874$.

6. (a) The uniform distribution of deaths assumption is that for an integer value x and for $0 \le t \le 1$,

$$l_{x+t} = l_x - t \cdot d_x$$

   (or, equivalently, $l_{x+t} = (1-t) \cdot l_x + t \cdot l_{x+1}$).

   (b) Consider ${}_{t+s}p_x = {}_{s}p_x \cdot {}_{t}p_{x+s}$. Therefore

$${}_{t}p_{x+s} = \frac{{}_{s+t}p_x}{{}_{s}p_x} = \frac{1 - {}_{s+t}q_x}{1 - {}_{s}q_x} = \frac{1-(s+t)\cdot q_x}{1 - s \cdot q_x},$$

   where the last equality follows by applying the UDD assumption to both the numerator and the denominator. This gives

$${}_{t}q_{x+s} = 1 - {}_{t}p_{x+s} = 1 - \left[\frac{1-(s+t)\cdot q_x}{1-s\cdot q_x}\right] = \frac{t \cdot q_x}{1 - s \cdot q_x}.$$

   (c) We want to evaluate ${}_{0.75}q_{60.75}$. Denoting this as ${}_{t}q_{x+s}$, then $t = 0.75$, $x = 60$ and $s = 0.75$. Therefore $s + t > 1$ and we cannot apply the result ${}_{t}q_{x+s} = \frac{t \cdot q_x}{1-s\cdot q_x}$ directly. Instead, tackle the problem by considering the probability over $(60.75, 61]$ and $[61, 61.5]$ separately, i.e. beginning with the survival probability to allow this separation,

$$\begin{aligned}
{}_{0.75}p_{60.75} &= {}_{0.25}p_{60.75} \cdot {}_{0.5}p_{61} = (1 - {}_{0.25}q_{60.75})\,(1 - {}_{0.5}q_{61}) \\
&= \left(1 - \frac{0.25 \times q_{60}}{1 - 0.75 \times q_{60}}\right)(1 - 0.5 \cdot q_{61}) \\
&= \left(1 - \frac{0.25(0.002451)}{1 - 0.75(0.002451)}\right)(1 - 0.5(0.002955)) = 0.9979095.
\end{aligned}$$

   Therefore ${}_{0.75}q_{60.75} = 0.00209$.

7. (i) For $l_{56.6}$, we have $x = 56$ and $t = 0.6$. Assuming UDD between integer ages 56 and 57,

$$l_{56.6} = 0.4(l_{56}) + 0.6(l_{57}) = 0.4(90490) + 0.6(89684) = 90006.4.$$

(ii) For ${}_{0.4}q_{60}$ we note that $x = 60$ and $t = 0.4$. Assuming UDD between integer ages 60 and 61,

$${}_{0.4}q_{60} = 0.4 \times q_{60} = 0.4(0.01392) = 0.005568.$$

(iii) ${}_3p_{45.25}$. Assuming UDD between integer ages

$${}_3p_{45.25} = \frac{l_{48.25}}{l_{45.25}} = \frac{l_{48} - 0.25d_{48}}{l_{45} - 0.25d_{45}} = \frac{94668 - 0.25 \cdot 352}{95521 - 0.25 \cdot 255} = 0.9908100.$$

(iv) For $\mu_{64.2}$, we note that $x = 64$ and $t = 0.2$. Assuming UDD between integer ages 64 and 65,

$$\mu_{64.2} = \frac{q_{64}}{1 - 0.2 \times q_{64}} = \frac{0.02199}{1 - 0.2(0.02199)} = 0.022087.$$

8. For $0 \le t \le 2$, the assumption is that $l_{60+t} = a + bt + ct^2$ for unknown constants a, b and c.

If $t = 0$, $\quad l_{60} = a = 86714$,

if $t = 1$, $\quad l_{61} = a + b + c = 85507$,

and if $t = 2$, $\quad l_{62} = a + 2b + 4c = 84173$.

Solving for a, b and c we get $l_{60+t} = 86714 - 1143.5t - 63.5t^2$.

Finally,

$${}_{0.4}q_{60} = 1 - {}_{0.4}p_{60} = 1 - \frac{l_{60.4}}{l_{60}} = 1 - \frac{86714 - 1143.5 \cdot 0.4 - 63.5(0.4)^2}{86714} = 0.0053920.$$

9. (a) The probability that a life currently age 75, who has just been selected, will survive to age 85 is

$${}_{10}p_{[75]} = \frac{\ell_{85}}{\ell_{[75]}} = \frac{10\,542}{15\,930} = 0.66177.$$

(b) The probability that a life currently age 76, who was selected one year ago, will die between ages 85 and 87 is

$${}_9|_2q_{[75]+1} = {}_9p_{[75]+1}{}_2q_{85} = \frac{\ell_{85} - \ell_{87}}{\ell_{[75]+1}} = \frac{10\,542 - 9\,064}{15\,668}\left(1 - \frac{9\,064}{10\,542}\right) = 0.09433.$$

(c)

$${}_4|_2q_{[77]+1} = \frac{12\,576 - 11\,250}{14\,744} = 0.08993.$$

10. (a)

$${}_2p_{[72]} = \left(1 - q_{[72]}\right)\left(1 - q_{[72]+1}\right) = (1 - 0.005236)\,(1 - 0.007456) = 0.98735.$$

(b)

$$\begin{aligned}
{}_3q_{[73]+2} =& q_{[73]+2} + p_{[73]+2}q_{[73]+3} + p_{[73]+2}p_{[73]+3}q_{[73]+4} \\
=& 0.011370 + (1 - 0.011370)\,(0.014988) \\
& \quad + (1 - 0.011370)\,(1 - 0.014988)\,(0.019316) \\
=& 0.04500.
\end{aligned}$$

(c)

$${}_1|q_{[65]+4} = p_{[65]+4}q_{[65]+5} = p_{[65]+4}q_{70} = (1 - 0.007994)\,(0.010599) = 0.01051.$$

(d)

$$\begin{aligned}
{}_7p_{[70]} &= \left(1 - q_{[70]}\right)\left(1 - q_{[70]+1}\right)\left(1 - q_{[70]+2}\right)\left(1 - q_{[70]+3}\right)\left(1 - q_{[70]+4}\right)(1 - q_{75})\,(1 - q_{76}) \\
&= (1 - 0.004285)\,(1 - 0.005967)\,(1 - 0.008066)\,(1 - 0.010629)\,(1 - 0.013698) \\
&\quad \times (1 - 0.018774)\,(1 - 0.021053) \\
&= 0.92027.
\end{aligned}$$